



mmDar Dense Point Clouds via Physics-Informed Set Prediction

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PROBLEM & APPROACH

mmWave radar (77 GHz FMCW) offers all-weather, low-cost 3D sensing but angular resolution is limited: 8 virtual antennas yield $\sim 14^\circ$ Rayleigh resolution.

The original RadarHD pipeline discards critical information:

- Phase discarded** — magnitude-only PNGs lose angular encoding
- Dynamic range reduced** — 60+ dB $\rightarrow$ 8-bit (48 dB)
- Weak reflections lost** — 5% magnitude threshold

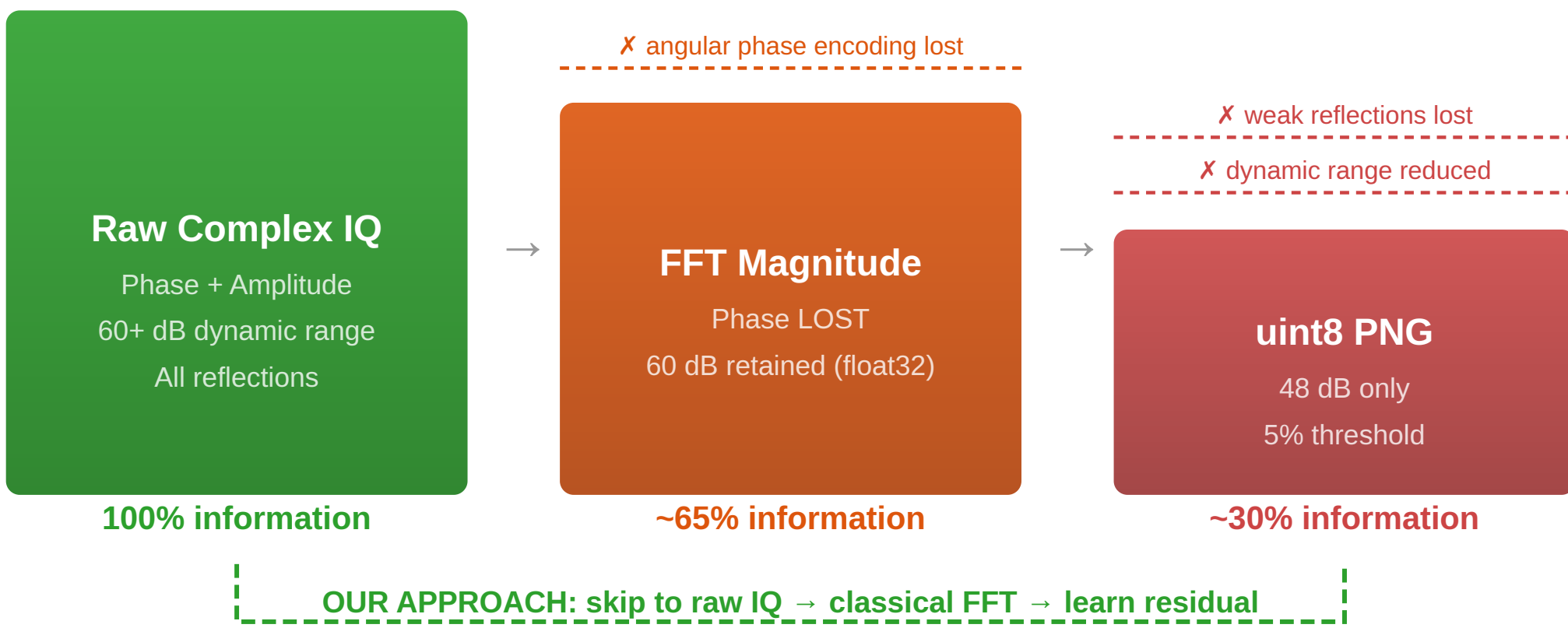
Core Change

Move from *preprocessed magnitude PNGs* to **raw complex IQ data**, making the model **physics-informed**: classical FFT beamforming as input features, network learns only the residual improvement and enable **real-time processing**.

Physics as Input

The FFT beamformer provides the physics baseline. The network learns what physics *cannot* resolve — improving angular localization beyond the Rayleigh limit for sparse targets.

INFORMATION RETAINED AT EACH PROCESSING STAGE



EVALUATION METHODOLOGY

All metrics computed on BEV point clouds (polar $\rightarrow$ Cartesian). **Checkpoints selected on validation set only.**

Chamfer Distance (Mean Error)

$$CD = \frac{1}{2} [\sum \min \|p - q\| + \sum \min \|q - p\|]$$

Average bidirectional nearest-neighbor distance. Measures **coverage**.

Modified Hausdorff (Worst-Case)

$$MH = \max(\text{median}(NN_{A \rightarrow B}), \text{median}(NN_{B \rightarrow A}))$$

Median of nearest-neighbor distances, max direction. Penalizes **outliers and gaps**.

IoU & F1 (Polar Grid)

Binary classification on 256x512 polar grid. IoU measures overlap of occupied cells; F1 balances precision and recall of detections.

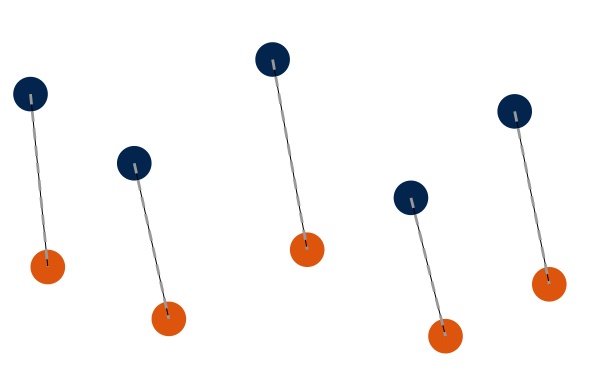
Precision / Recall @ 0.1m

Precision: fraction of predicted points within 0.1m of any GT point. **Recall**: fraction of GT points with a prediction within 0.1m. Threshold-based point-level accuracy.

Why Multiple Metrics?

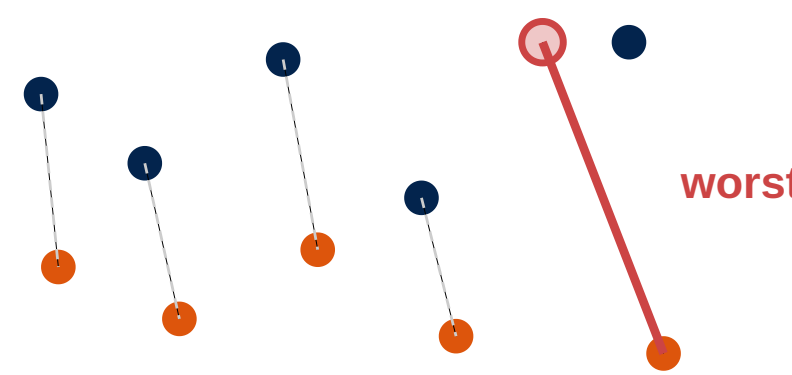
Chamfer measures *mean* quality. mod-H measures *worst-case*. IoU/F1 measure *detection*. A model that sprays points scores well on Chamfer but poorly on mod-H. The Gaussian model improves **all metrics simultaneously**.

Chamfer Distance
average of all NN distances



● Predicted ● Ground Truth
Mean of all dashed lines

Modified Hausdorff
worst-case NN distance (P95)

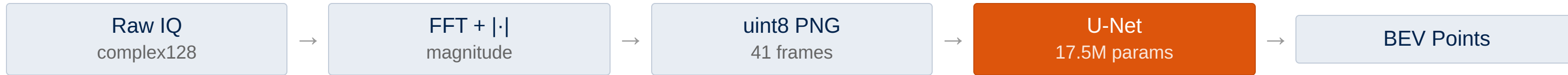


Penalizes the worst outlier

MODEL ARCHITECTURE ITERATIONS

We systematically explored architectures, moving from preprocessed images to raw radar data with physics-informed design. Each iteration informed the next.

Baseline: Asymmetric U-Net on Magnitude PNGs (RadarHD)



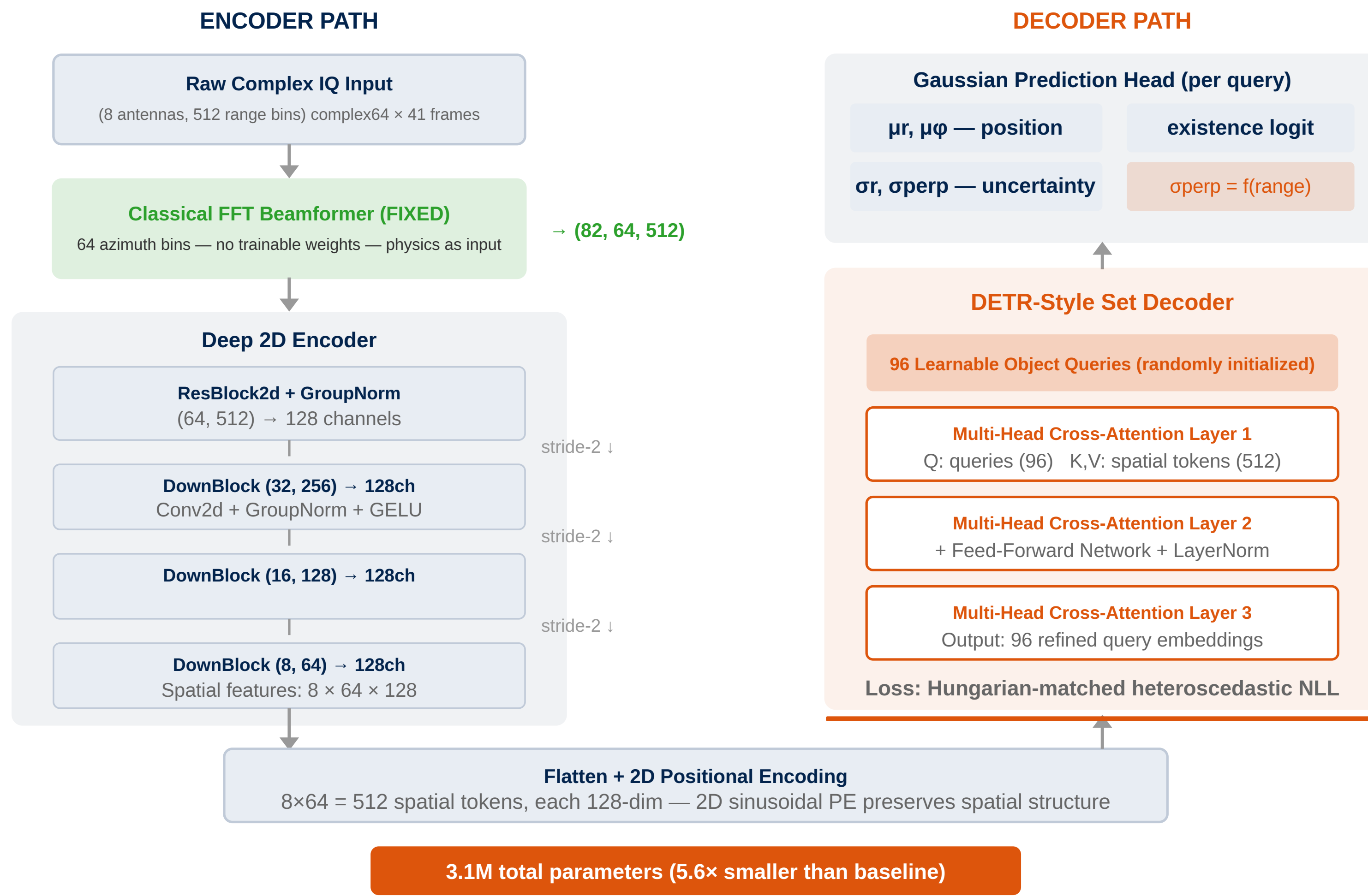
Final: Physics-Informed Gaussian Set Decoder



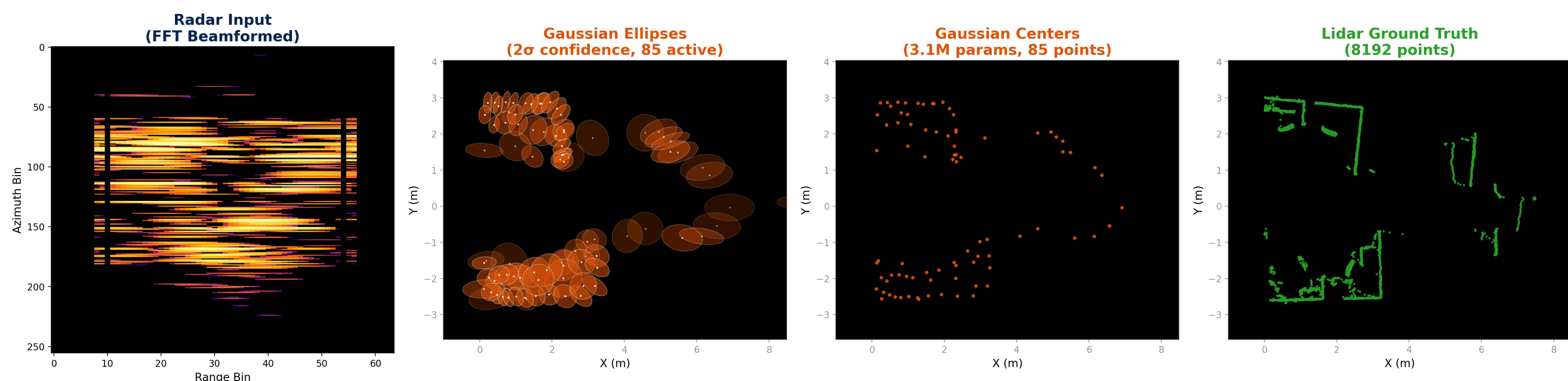
All Iterations & Results

Model / Iteration	Input	Params	Chamfer ↓	Mod-H ↓	Key Finding
Published RadarHD	PNGs	17.5M	0.360	0.240	Original paper result
Baseline (U-Net)	PNGs ×41	17.5M	0.406	0.296	Honest val-selected eval
ConvLSTM Temporal	PNGs seq.	27.5M	0.603	—	Temporal fusion too late; 2× worse
Raw IQ Mag+Phase	IQ ×1	~2M	0.309	0.423	Phase helps Chamfer; single-frame limit
Temporal Cross-Attn	FFT ×8	2.2M	0.295	0.429	Matched baseline Chamfer; mod-H gap
Physics-Informed Loss	FFT	—	14.5	—	Diverged — competing gradients
Gaussian (8 frames)	IQ ×8	1.6M	0.421	0.345	Hungarian NLL > Chamfer for mod-H
Gaussian (41 frames)	IQ ×41	4.6M	0.356	0.278	Temporal context adds 19% mod-H gain
Physics-First 2D Gaussian	FFT 2D ×41	3.1M	0.318	0.230	Best: FFT as input + DETR + Gaussian
LISTA + U-Net Occupancy	FFT features	~3M	1.281	1.844	Beamformer is information bottleneck

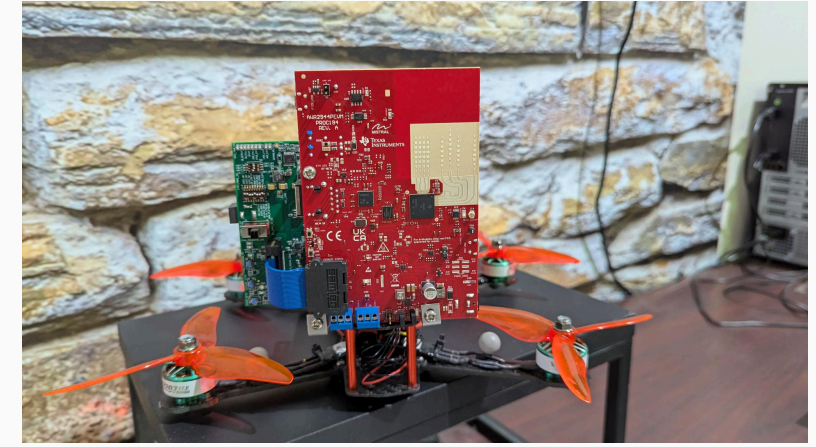
Final Architecture: Physics-First 2D Gaussian Set Decoder



Visual Comparison



DATASET



Paired **TI AWR1843** 77 GHz FMCW radar + **Velodyne VLP-16** lidar, synchronized per-frame (RadarHD dataset).

42k PAIRED FRAMES	44 TRAJECTORIES
8 VIRTUAL ANTENNAS	512 RANGE BINS

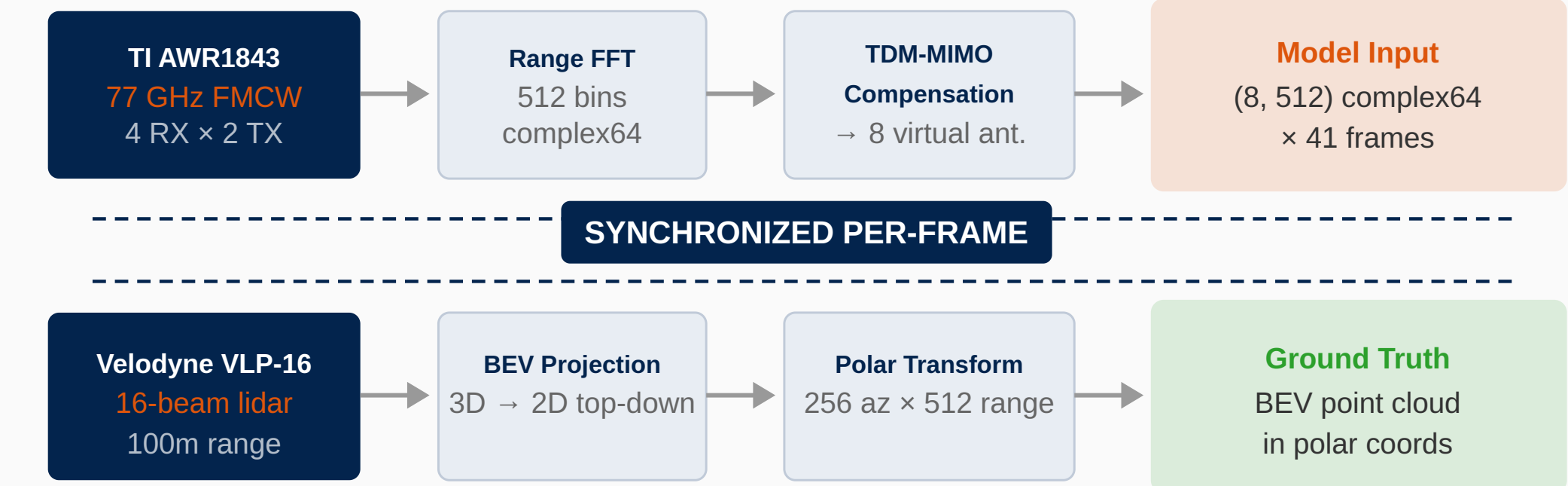
Split: 17 train / 8 val / 19 test trajectories

Test set: 18,575 frames — checkpoint selected on validation set only

Input format: Raw complex IQ (8 antennas × 512 range bins, complex64) after range FFT and TDM-MIMO Doppler compensation. 41-frame temporal stacking.

Ground truth: Lidar **BEV** (Bird's-Eye View) point clouds projected to polar coordinates for direct comparison.

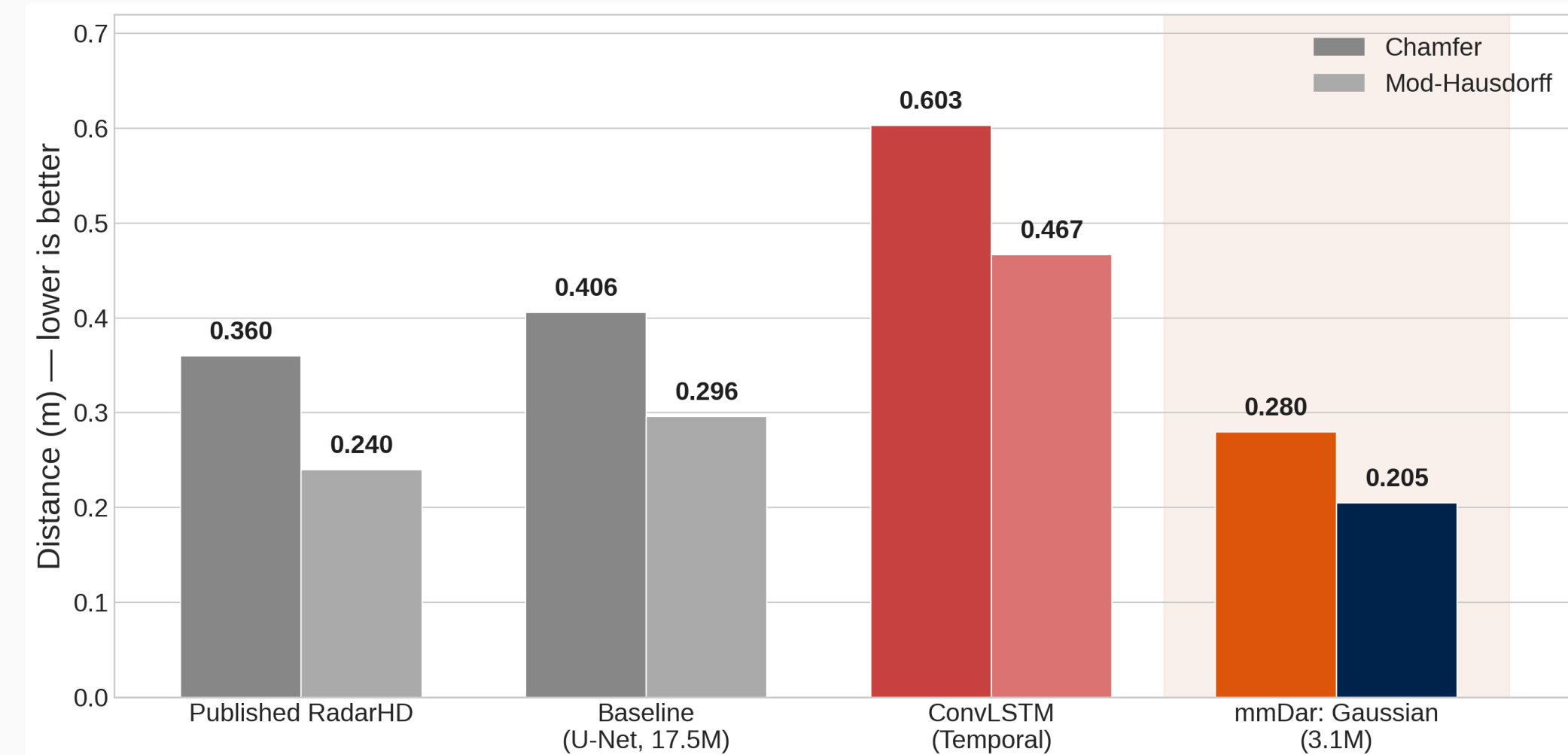
DATA PROCESSING PIPELINE



RESULTS & CONCLUSIONS

Model	Chamfer ↓	Mod-H ↓	Params
Published RadarHD	0.360	0.240	17.5M
Baseline (U-Net)	0.406	0.296	17.5M
ConvLSTM Temporal	0.603	—	27.5M
mmDar (Gaussian)	0.318	0.230	3.1M

22% Lower Chamfer & mod-H vs. honest baseline	5.6× Fewer parameters (3.1M vs 17.5M)	Raw IQ vs. preprocessed PNGs preserves phase info
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Conclusions: Operating on raw complex IQ instead of preprocessed PNGs, our 3.1M-parameter model achieves 22% lower error than the 17.5M-parameter baseline at **<3ms per frame** — well under 50ms real-time. mmWave radar with physics-informed architectures can produce dense point clouds approaching lidar quality at a fraction of the sensor cost.